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ERRATUM: "HABITABLE ZONES AROUND MAIN-SEQUENCE STARS: NEW ESTIMATES" (2013, ApJ, 765, 131)

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Online-only material: color figures

Due to an error at the publisher, several changes and corrections made at the proof stage were not included in the published version of the paper. The final results are not affected significantly. The coefficients used to calculate the habitable zone (HZ) limits have changed slightly (∼1.5%). Accordingly, we have provided the updated values here, and also updated the online HZ calculator[10,11] and the FORTRAN code available online.

1. In Section 2.1 "Model Updates," point number 2, the Rayleigh scattering by water vapor should be as follows, following a correction noted by Von Paris et al. (2010):

$$\sigma_{\mathrm{R,H_2O}}(\lambda) = 4.577 \times 10^{-21}\left(\frac{6+3D}{6-7D}\right)\left(\frac{r^2}{\lambda^4}\right)\mathrm{cm}^2. \tag{1}$$

   Here, D is the depolarization ratio (0.17 for H_2O; Marshall & Smith 1990), r is the wavelength (λ)-dependent refractivity which is calculated as $r = 0.85\, r_{\mathrm{dryair}}$ (Edlén 1996), r_{dryair} is obtained from Equation (4) of Bucholtz (1995), and λ is in microns.
   Furthermore, the paragraph following this equation that starts with "Note that the coefficient..." should be discarded as Von Paris et al. (2010) published an erratum correcting their Rayleigh scattering expression.
   Also, in the same paragraph, the value 2.5×10^{-27} cm^2 should be 2.6×10^{-27} cm^{-2}.
2. In footnote 11, Section 1 ("Introduction"), the following sentence should be inserted after ". . . .much closer to their host stars": "However, Abbot et al. (2012) found that a waterworld would have a narrower HZ owing to lack of weathering-climate feedback."
3. Figure 1 in the published version should be replaced with the figure shown here. The OLR from our model increased slightly from 86 Wm^{-2} to 88 Wm^{-2}.
4. Figure 3(b) should be replaced by the figure shown here.
5. In Section 3.3, last paragraph, the sentence ". . .3 bar atmosphere containing 90% CO_2 and 10% H_2 could have. . ." should read as "∼1.5 bar atmosphere containing 80% CO_2 and 20% H_2 could have. . . ."
6. In Section 4.1, the sentence ". . .and metallicities ([Fe/H] 4.0 to +0.5 needed to simulate stellar spectra." should read as ". . .and metallicity ([Fe/H] = 0.0) needed to simulate stellar spectra."
7. In the paragraph preceding Equation (4), the following papers should be cited after the sentence ". . . .habitability over an eccentric orbit": (Kopparapu et al. 2009; Kopparapu & Barnes 2010).
8. In the paragraph following Equation (4), the following sentence should be added after the sentence "This may help eccentric planets near the OHZ maintain habitable conditions."
   "However, obliquity variations can influence the geographical distribution of irradiation (Spiegel et al. 2008, 2009; Dressing et al. 2010) and may change habitable conditions."
9. In Table 1, the maximum greenhouse for the outer edge of the HZ should be 1.67 AU, instead of 1.70 AU. Therefore, Table 1 should be updated accordingly.
10. Figure 6(a) is updated with the that shown here.
11. The coefficients listed in Table 3 must be updated. The average change in the HZ boundaries from the published version is 0.5% at the inner edge and 3% at the outer edge. The corresponding FORTRAN code (included in this erratum) and the online HZ calculator are also updated.

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[10] http://www3.geosc.psu.edu/∼ruk15/planets/
[11] http://depts.washington.edu/naivpl/content/hz-calculator

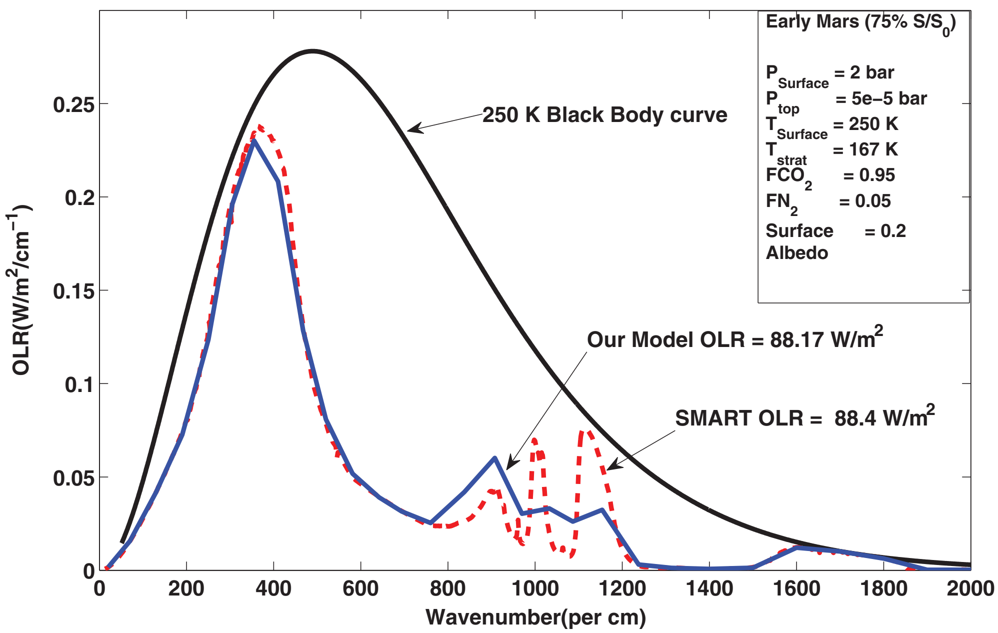

Figure 1. Plot of outgoing long-wave radiation vs. wavenumber for the 0–2000 cm^{-1} region comparing our OLR (blue solid curve) to that from SMART (red dashed curve). This calculation is for early Mars conditions, 2 bar CO_2, and constant stratospheric and surface temperatures of 167 and 250 K, respectively. The corresponding 250 K blackbody curve is shown in black. The integrated flux over all bands at the top of the atmosphere is 88.17 Wm^{-2} for our model and 88.4 Wm^{-2} for SMART.

(A color version of this figure is available in the online journal.)

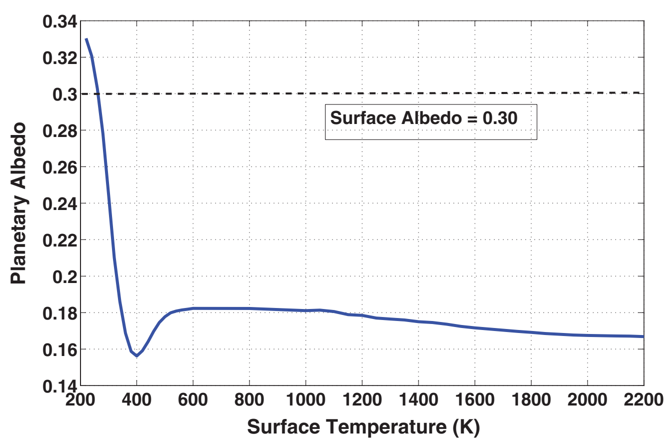

Figure 3. Inner edge of the habitable zone calculations from our updated climate model. Various parameters are shown as a function of surface temperature: (b) planetary albedo.

(A color version of this figure is available in the online journal.)

Table 1
Habitable Zone Distances around Our Sun from Our Updated 1D Climate Model

Model	Inner Habitable Zone			Outer Habitable Zone	
	Moist Greenhouse	Runaway Greenhouse	Recent Venus	Maximum Greenhouse	Early Mars
This paper	0.99 AU	0.97 AU	0.75 AU	1.67 AU	1.77 AU
Kasting et al. (1993)	0.95 AU	0.84 AU	0.75 AU	1.67 AU	1.77 AU

Note. For comparison, estimates from Kasting et al. (1993) are also shown.

12. The acknowledgements should be expanded as follows.

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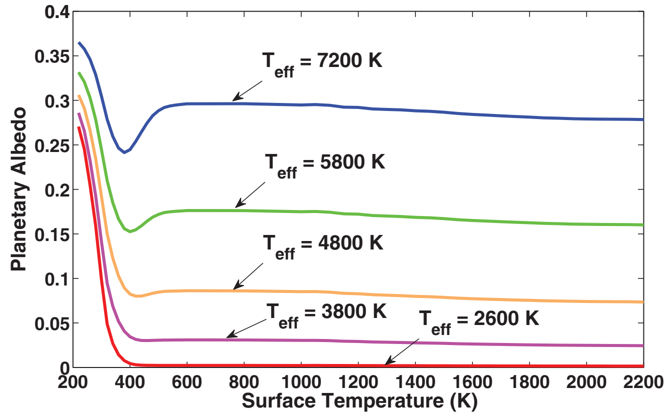

Figure 6. Habitable zone calculations from our climate model for stellar effective temperatures corrresponding to F ($T_{\rm eff} = 7200$ K), G (Sun), K ($T_{\rm eff} = 4800$ K), and M ($T_{\rm eff} = 3800$ K, and 2600 K) spectral types. The inner edge results are shown in (a).

(A color version of this figure is available in the online journal.)

Table 3
Updated Coefficients to Calculate Habitable Stellar Fluxes, and Corresponding Habitable Zones, for Stars with $2600 \leqslant T_{\rm eff} \leqslant 7200$ K

Constant	Recent Venus	Runaway Greenhouse	Moist Greenhouse	Maximum Greenhouse	Early Mars
$S_{\rm eff\odot}$	1.7763	1.0385	1.0146	0.3507	0.3207
a	1.4335×10^{-4}	1.2456×10^{-4}	8.1884×10^{-5}	5.9578×10^{-5}	5.4471×10^{-5}
b	3.3954×10^{-9}	1.4612×10^{-8}	1.9394×10^{-9}	1.6707×10^{-9}	1.5275×10^{-9}
c	-7.6364×10^{-12}	-7.6345×10^{-12}	-4.3618×10^{-12}	-3.0058×10^{-12}	-2.1709×10^{-12}
d	-1.1950×10^{-15}	-1.7511×10^{-15}	-6.8260×10^{-16}	-5.1925×10^{-16}	-3.8282×10^{-16}

Note. An ASCII file containing these coefficients can be downloaded in the electronic version of the journal.

computing resources and services that have contributed to the research results reported in this paper (http://rcc.its.psu.edu). This work was also facilitated through the use of advanced computational, storage, and networking infrastructure provided by the Hyak supercomputer system, supported in part by the University of Washington eScience Institute. This research has made use of the Exoplanet Orbit Database and the Exoplanet Data Explorer at http://exoplanets.org.

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IOP Publishing sincerely regrets these errors.

Note added in proof. R. Wordsworth pointed out that his model uses increased vertical resolution in the lower atmosphere and that his results were found to be insensitive to further increases in resolution. All such calculations should be tested to determine whether they are robust to this issue.

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